Dealer Networks: Cocaine vs. Heroin Networks of the 90s

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Why Use Two Networks?

Two Moments.

Two Markets.

One Story.

Networks Used: New York City Wiretaps

- Heroin network – 1990
- Cocaine network – 2000

What Two Networks Reveal:

- How drug types shape dealer networks
- Unintended structural consequences of the opioid crisis

Basics of the Network:

01

Heroin Network Nodes

38 Nodes
Heroin Dealers

02

Heroin Network Ties

Dealers are connected if they exchanged information

- Undirected
- Unweighted

03

Cocaine Network Nodes

28 Nodes
Cocaine Dealers

04

Cocaine Network Ties

Dealers are connected if they exchanged information

- Directed
- Weighted

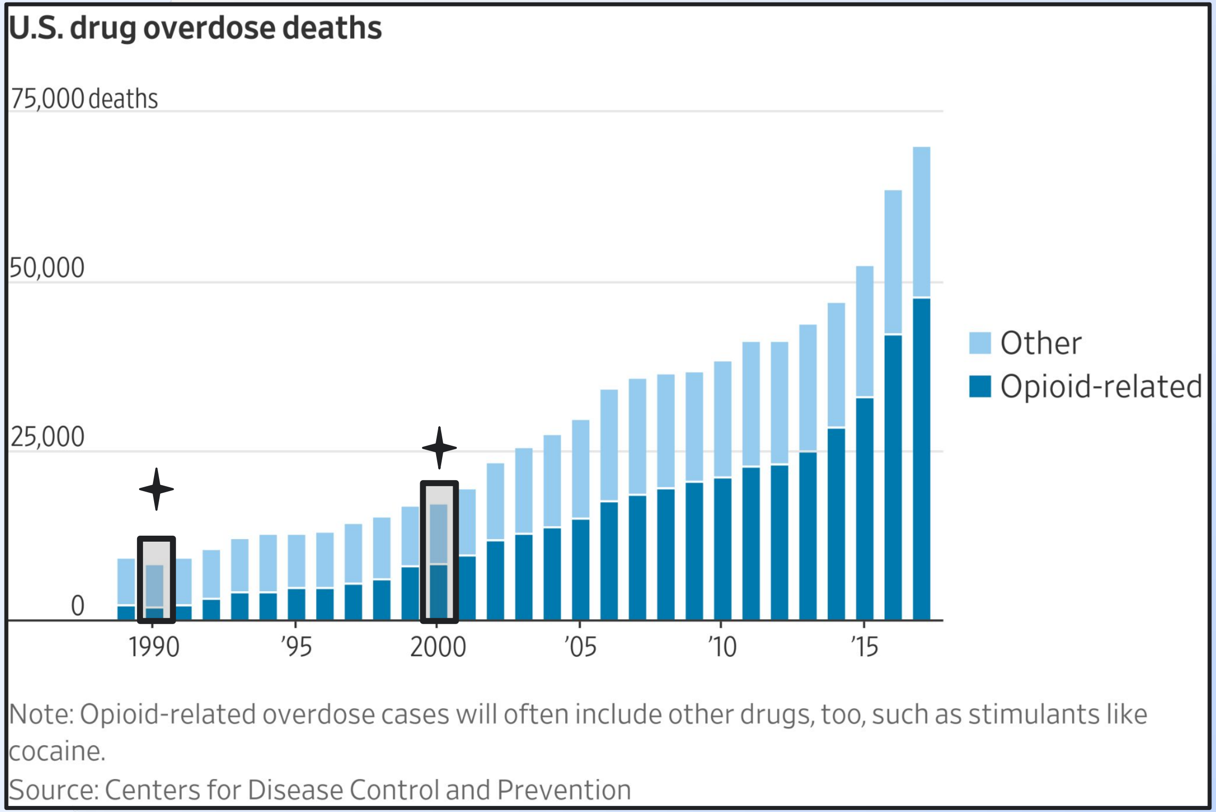
Unintended Discovery:
Capturing the Rise of the Opioid Epidemic

Heroin Network captured a pre-opioid period.

Cocaine Network captures the turning point as the first wave hit.

Together they create a snapshots of an unfolding epidemic.

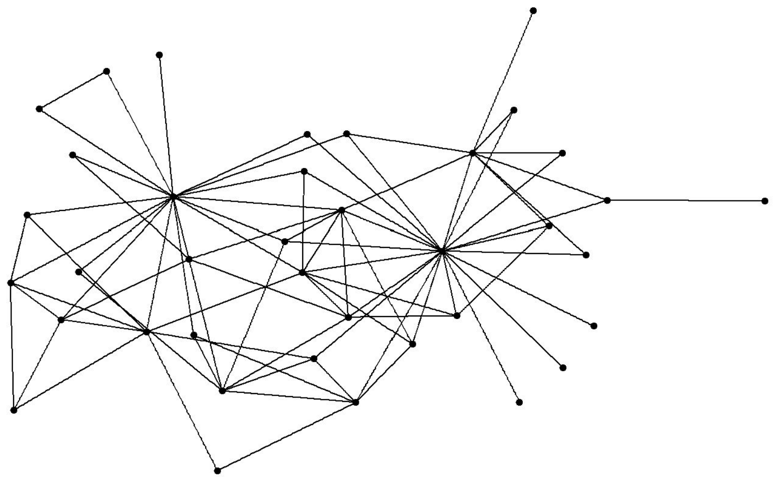
**Note cocaine is the most common addition in opioid-related deaths*



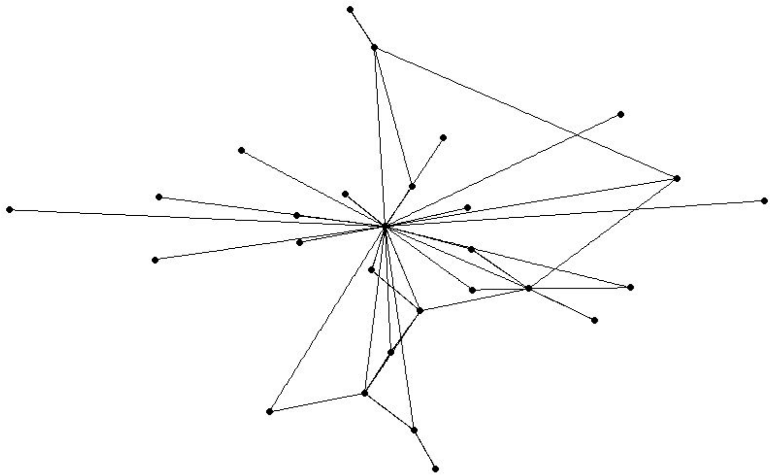
First Look at Both Networks

**Using stress models*

Heroin Network



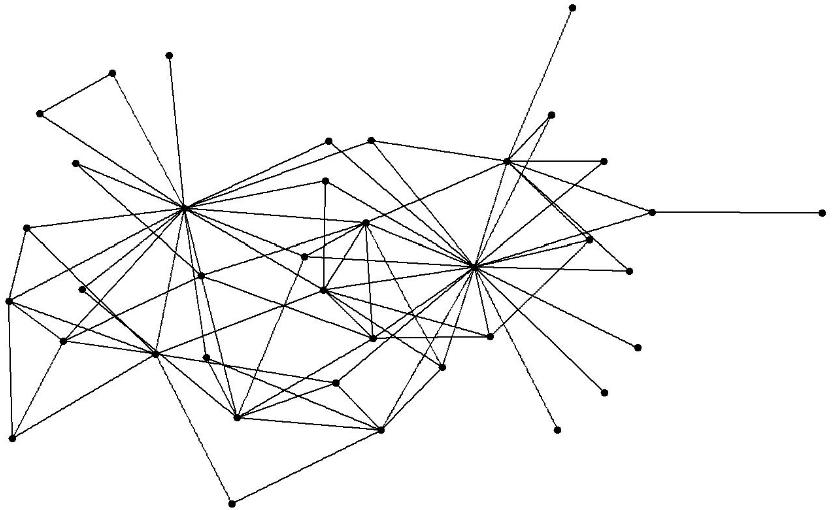
Cocaine Network



Analyzing Initial Structures

Heroin Network

Heroin network



Multiple central actors rather than a single dominant hub

High cross cluster connectivity

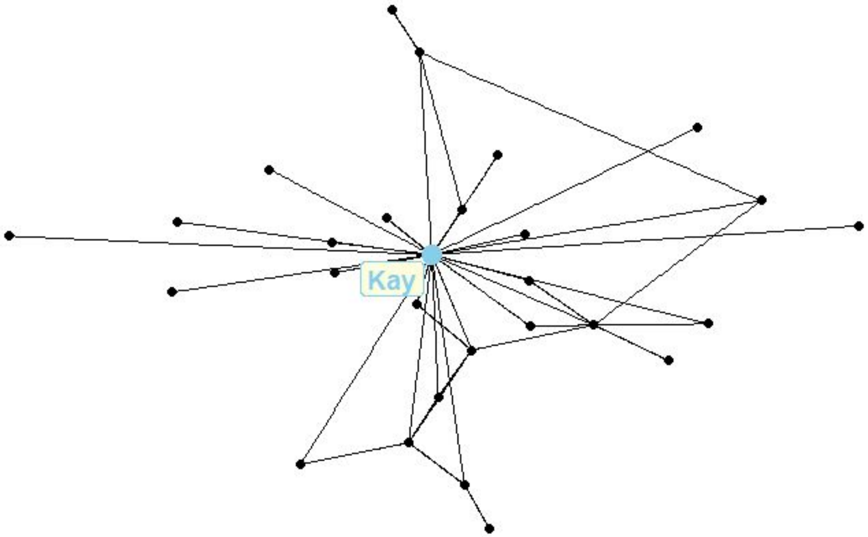
Analyzing Initial Structures

Cocaine Networks

Single central actor, Kay, becomes a person of interest as analysis continues

Strong Hierarchical Structures

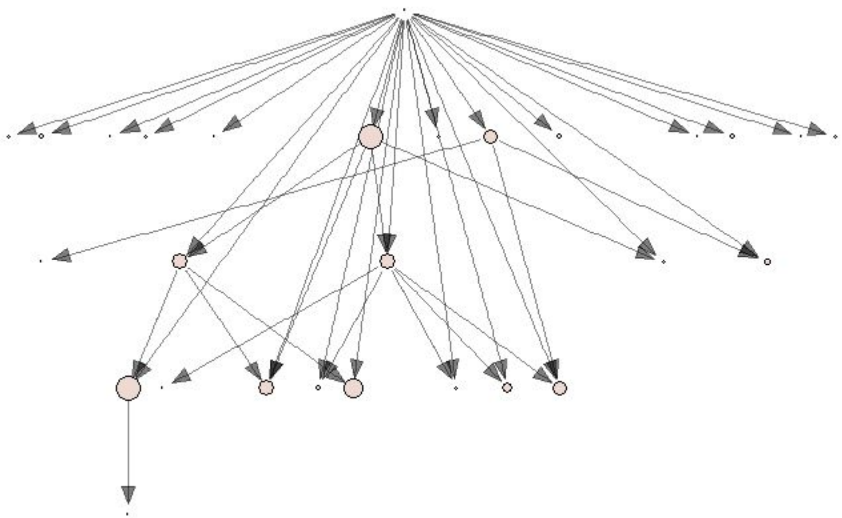
Cocaine Network



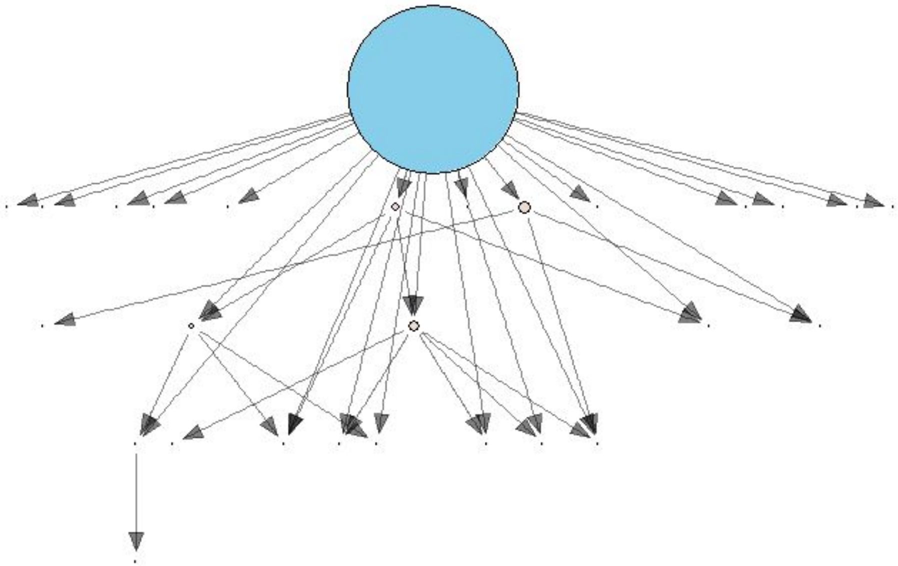
Tracking Movement

Cocaine Network

Graph Strength – In



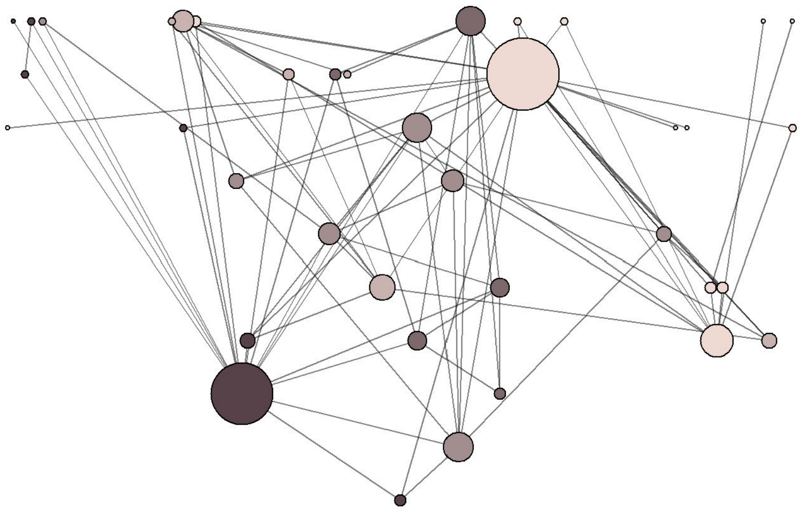
Graph Strength – Out



Complex Connections Reveal Cluster Structure

Heroin Network

Heroin network



Heroin dealer network breaks into clear groups

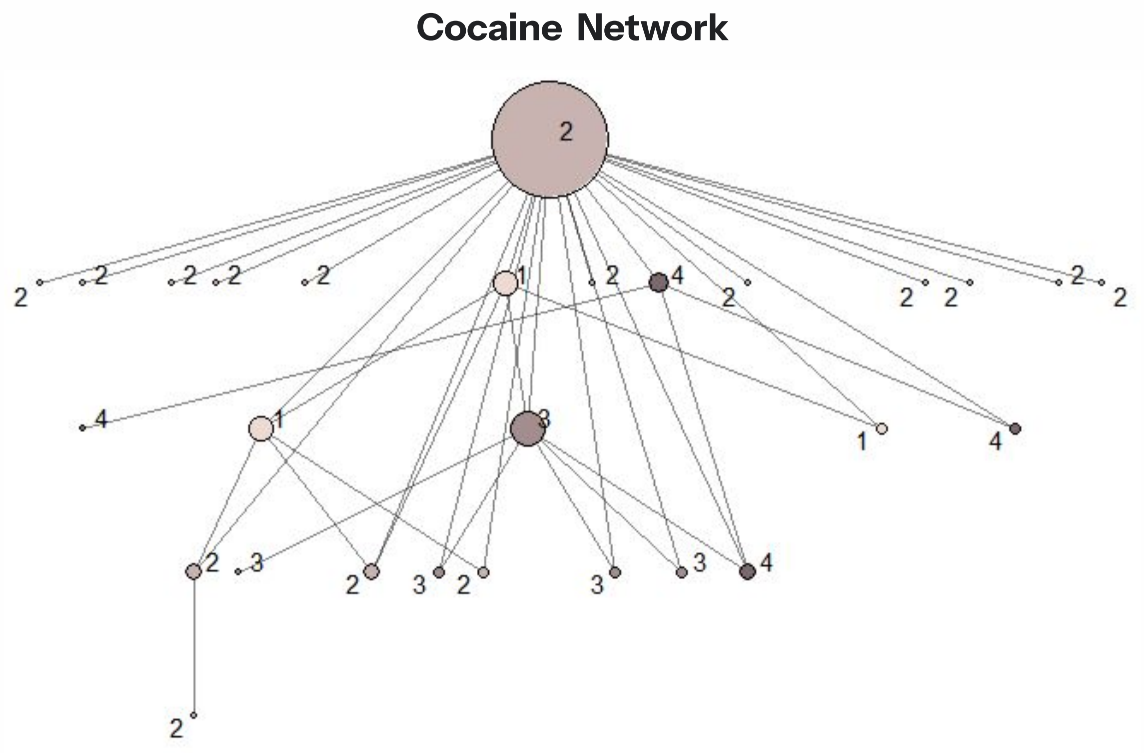
Some nodes act as connectors between groups
-Louvain Method

Hierarchical Organization Obscures Clusters

Cocaine Network

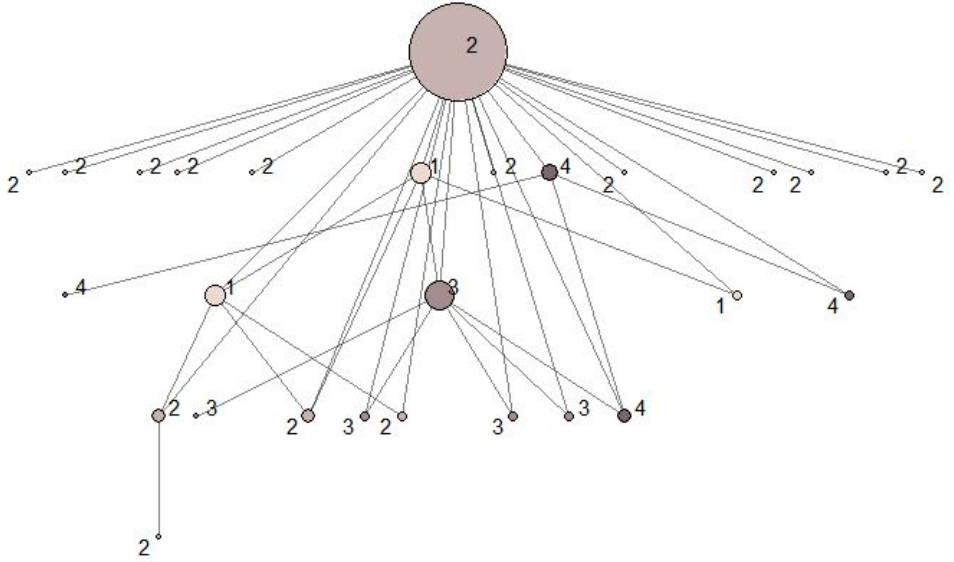
Custom Spinglass

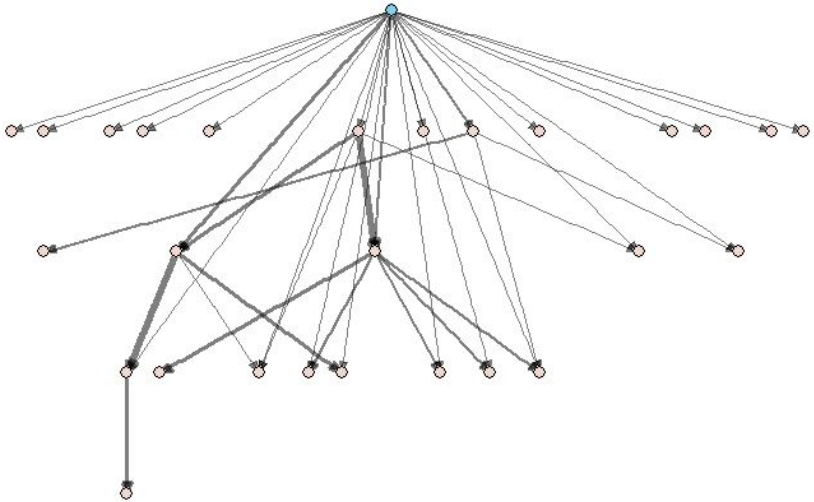
At first glance clusters appear random



Hierarchical Organization Obscures Clusters

Cocaine Network





Appears that most clustering algorithms favored edge weight

Understanding the Imbalance of Information

Information Centrality

Heroin Network

Min 0.011	The least central node still retains some access to information within the network
Median 0.0264	On average, actors have moderate access to information flows across the network
Max 0.041	The most central actor is more than 2x as embedded in information pathways as the least central actor

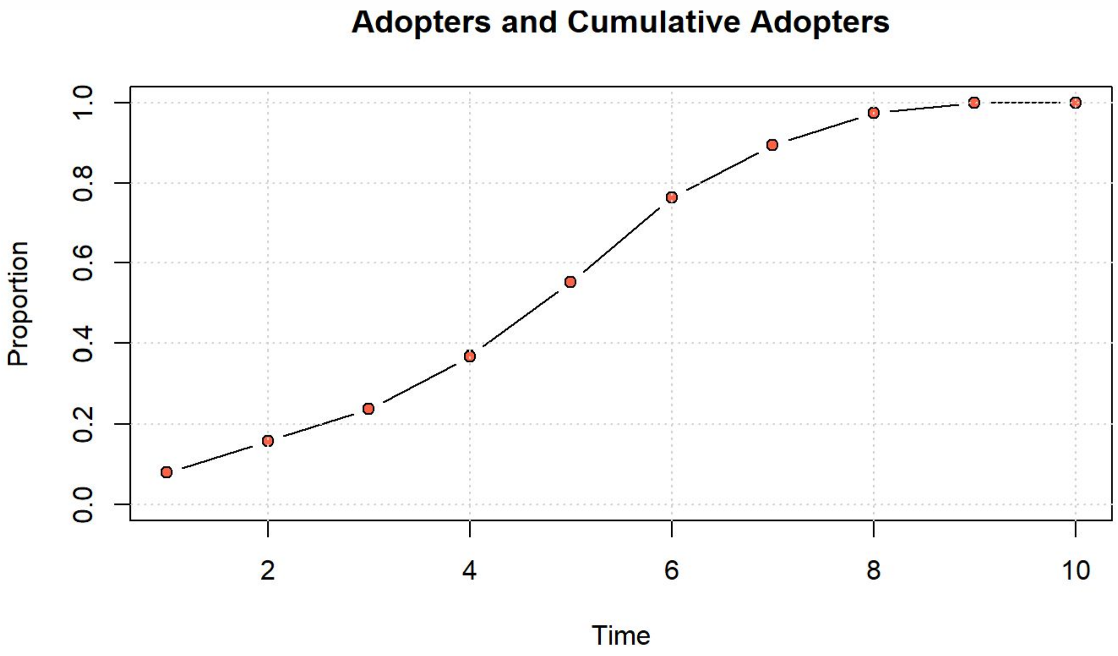
Cocaine Network

Min 0.022	Peripheral street-level dealers with minimal information control
Median 0.033	Typical Dealer likely mid-level control
Max 0.063	Most Central Dealer holds **3x more information** than lowest level street dealer

All values rescaled

Fractured Information Slows Diffusion

Heroin Network



Adoption Rate = 20%

Gradual adoption over time, reaching 0.55 cumulative adoption by period 5

Total adoption by period 9

Adoption spreads through connected clusters rather than randomly

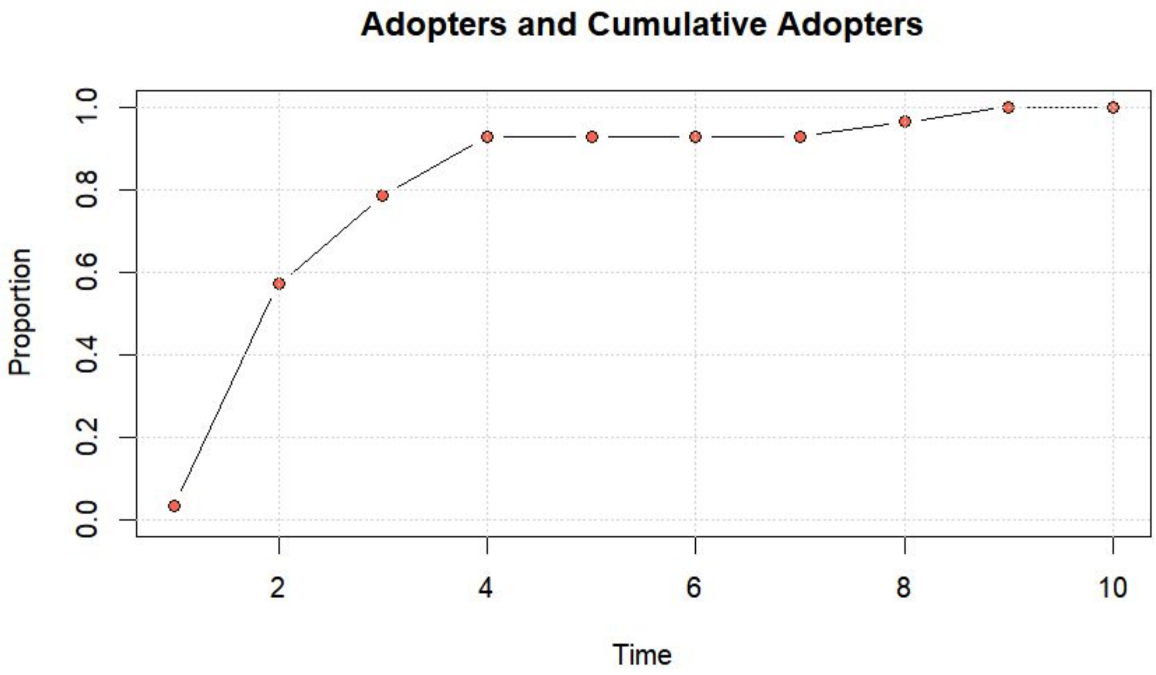
Centralized Information Increases Initial Diffusion

Cocaine Networks

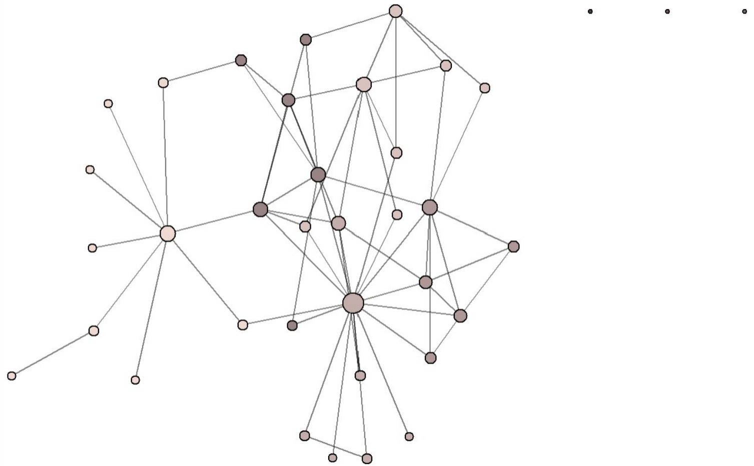
Adoption accelerates rapidly over short periods of time, reaching 0.96 cumulative adoption by period 5

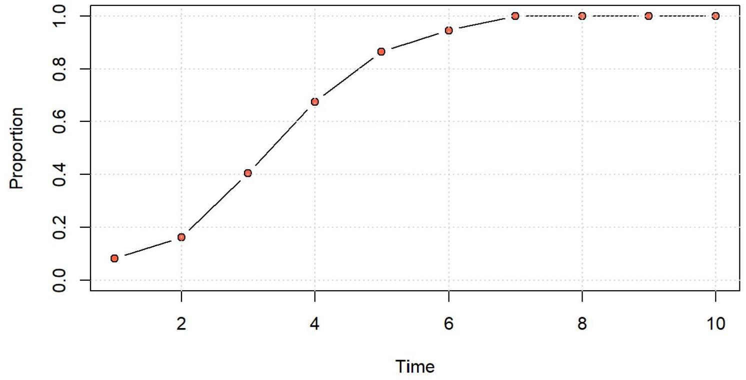
Total adoption by period 10

Diffusion moves top down, reaching mass market quickly then slowing as dealers reach peripheral nodes

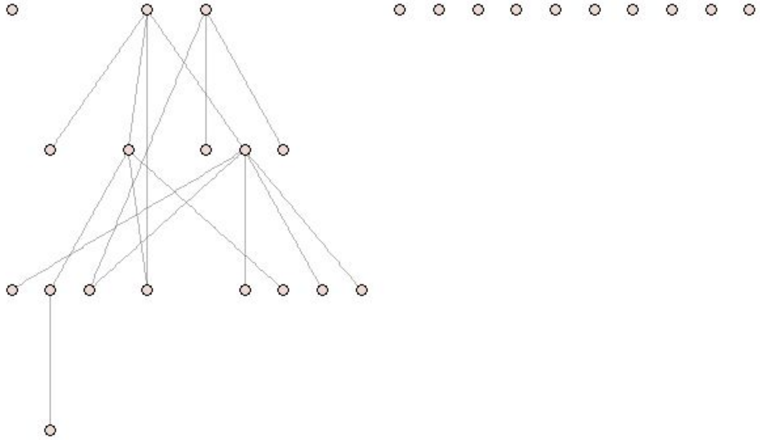


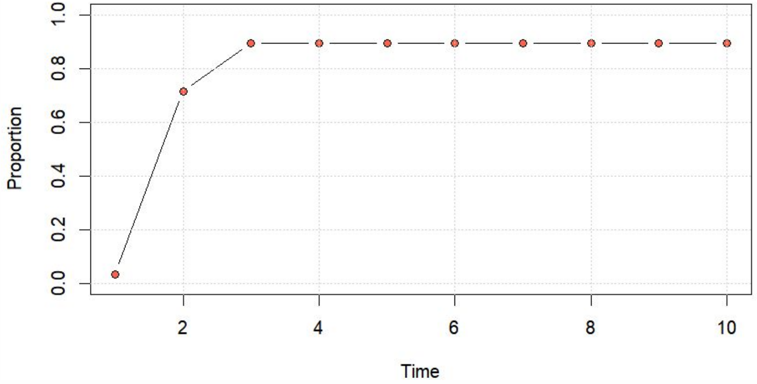
Heroin Network: Most Central Node Removed





Cocaine Network: Most Central Node Removed





Back to the Big Picture

Two Moments.

Two Markets.

One Story.

Heroin Network

- Complex network in a fractured market space
- Multiple points of strong informational centrality creates a network that is is more resistant to failure
- Inter connected clustered increased local efficiency
- Steady diffusion

Back to the Big Picture

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Cocaine Network

- Strong hierarchical structure leads to a power imbalance
- Single points of informational centrality leaves the network more susceptible to failure
- Enables rapid diffusion to a majority of the network

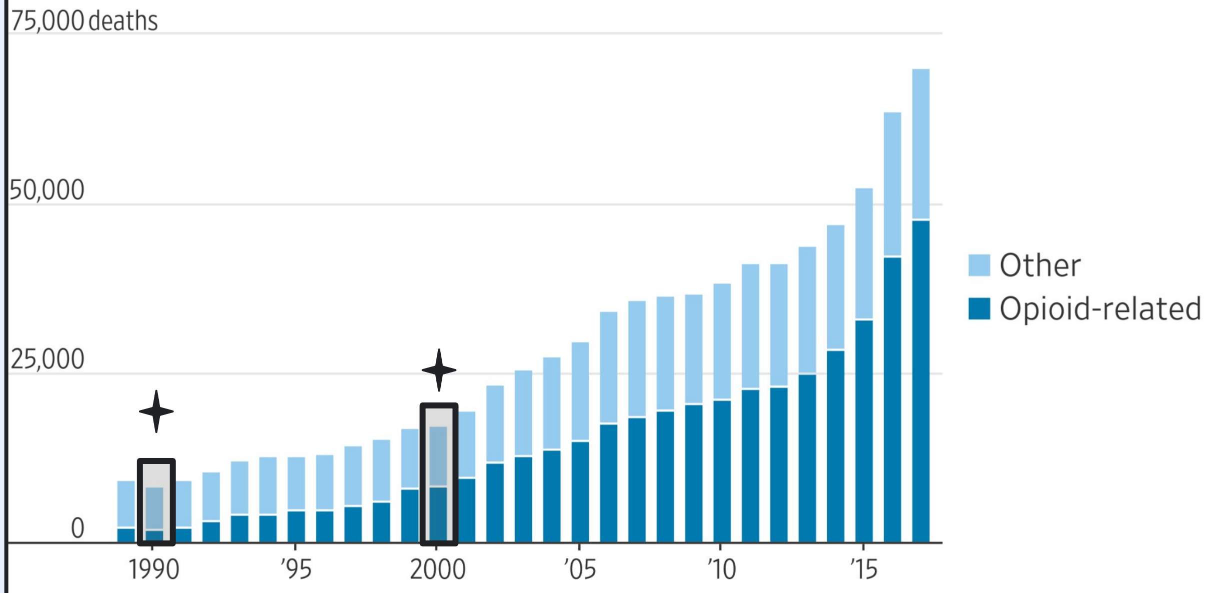
Bringing the Opioid Crisis Back into Context

While there are fundamental differences to these networks there may be a story that transcends both.

The heroin network tells of a time before mass crisis where diffusion is slow and steady.

The cocaine network captures the birth of the greatest public health crisis to date. With a centralized structure diffusion acts rapidly.

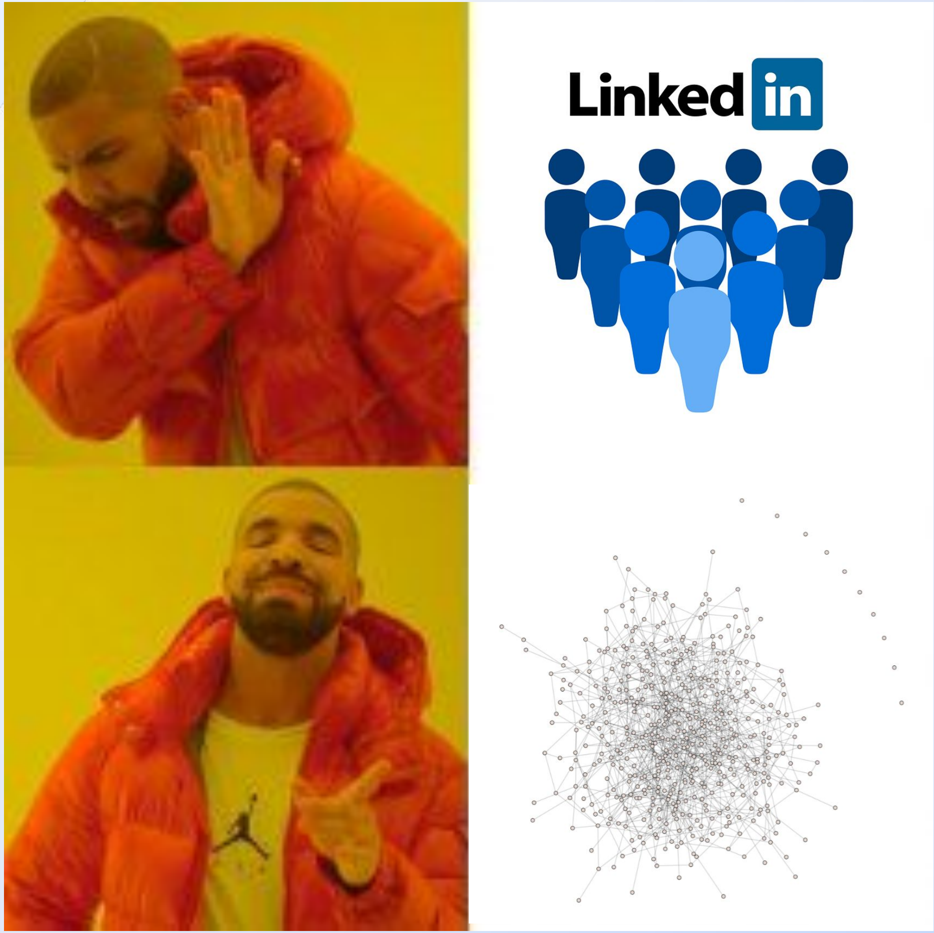
U.S. drug overdose deaths



Note: Opioid-related overdose cases will often include other drugs, too, such as stimulants like cocaine.
Source: Centers for Disease Control and Prevention

Thank you

Our career advisor keeps telling us…
“ABN: always be networking”



Unused Slides:

Noticeable Similarities and Differences

Heroin Network Centralization Stats

Degree Centralization: 0.42
Closeness Centralization: 0.34
Betweenness Centralization: 0.41
Eigenvector Centralization: 0.73

Cocaine Network Centralization Stats

Degree Centralization: 0.41
Closeness Centralization: 0.85
Betweenness Centralization: 0.01
Eigenvector Centralization: 0.81

Global Stats Comparison

Heroin Network

Average Local Efficiency: 25.4%

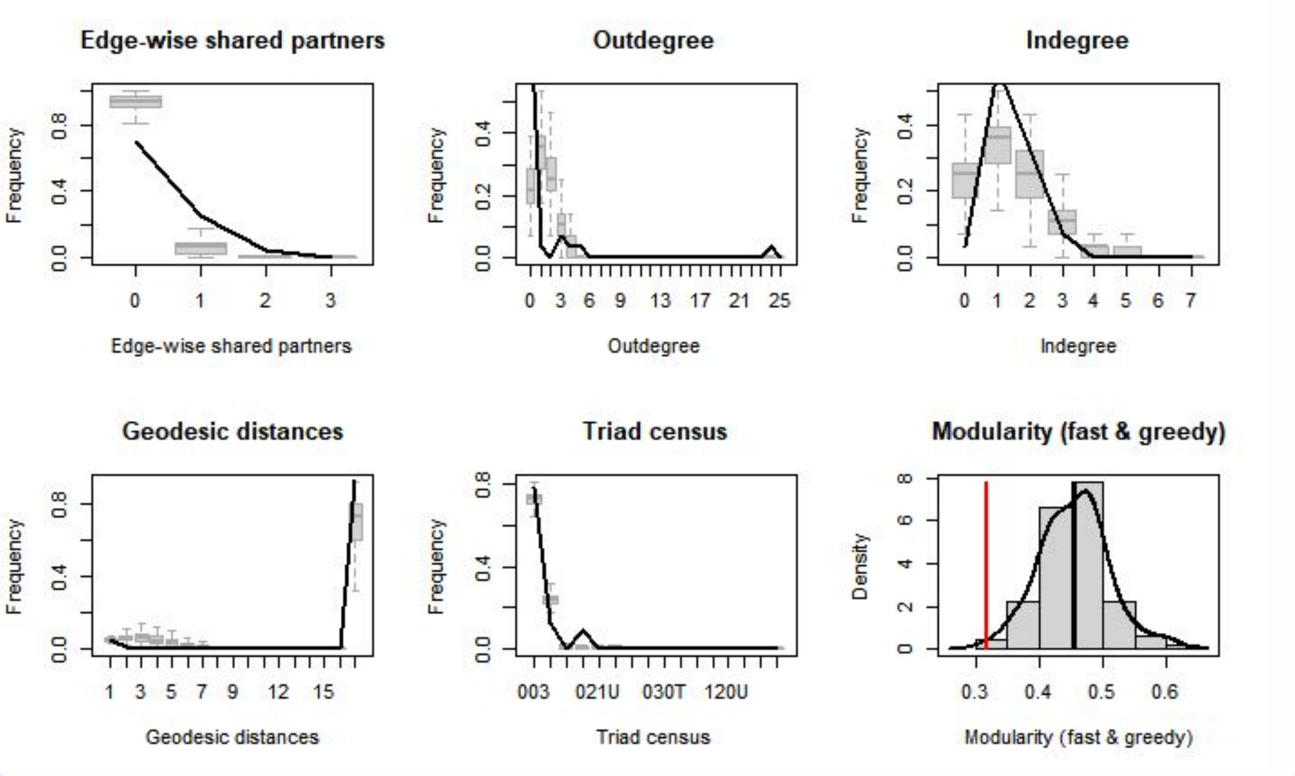
Mean Distance: 1.90

Cocaine Network

Average Local Efficiency: 2.8%

Mean Distance: 3.85

Further Analyzing Cocaine Networks - ERGM GOF



Add notes/sharpen text!
Given the limited data a base ERGM model has a strong GOF and high p-values, some individual components may be slightly off but model performs well.